Summary in Graph

Exam Summary (GO Classes GATE CS | Set Theory | Weekly Quiz 2)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	10 (5 + 5)
Total Marks:	15 (5 + 10)
Exam Duration:	30 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained

Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8 Q9 Q10

* Average Stats based on first-attempt stats of Top 10% (44) out of 439 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Set Theory & Algebra

If A and B are two sets and $A \cup B = A \cap B$ then

A. $A = \phi$
B. $B = \phi$
C. $A \neq B$
D. $A = B$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 Numerical Type | Award: 1 | Penalty: 0 | Set Theory & Algebra

The cardinality of the power set of $A \cup B$, where $A = \{2, 3, 5, 7\}$ and $B = \{2, 5, 8, 9\}$, is?

Your Answer: | Correct Answer: 64 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Set Theory & Algebra

What is the Cartesian product of $A = \{1, 2\}$ and $B = \{a, b\}$?

A. $\{(1, a), (1, b), (2, a), (b, b)\}$
B. $\{(1, 1), (2, 2), (a, a), (b, b)\}$
C. $\{(1, a), (2, a), (1, b), (2, b)\}$
D. $\{(1, 1), (a, a), (2, a), (1, b)\}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Set Theory & Algebra

Let $\mathcal{P}(A)$ denote the power set of A. If $\mathcal{P}(A) \subseteq B$ then

A. $2^{|A|} \leq |B|$
B. $2^{|A|} \geq |B|$
C. $2|A| < |B|$
D. $2^{|A|} \geq 2^{|B|}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Set Theory & Algebra

Let A be a finite set of size n. The number of elements in the power set of $A \times A$ is:

A. 2^{n^2}
B. $(2^n)^2$
C. $(2^2)^n$
D. None of the above

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 Multiple Select Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

Which one of the following is/are true?

A. $R \cap S = (R \cup S) - [(R - S) \cup (S - R)]$
B. $R \cup S = (R \cap S) - [(R - S) \cup (S - R)]$
C. $R \cap S = (R \cup S) - [(R - S) \cap (S - R)]$

D. $R \cap S = (R \cup S) \cup (R - S)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

Let A and B be sets in a finite universal set U. Given the following : $|A - B|, |A \oplus B|, |A| + |B|$, and $|A \cup B|$ Which of the following is in order of increasing size ?

A. $|A - B| < |A \oplus B| < |A| + |B| < |A \cup B|$
B. $|A \oplus B| < |A - B| < |A \cup B| < |A| + |B|$
C. $|A \oplus B| < |A| + |B| < |A - B| < |A \cup B|$
D. $|A - B| < |A \oplus B| < |A \cup B| < |A| + |B|$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 02sec | Discuss

Q #8 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

Let $A = \{0, 1\} \times \{0, 1\}$ and $B = \{a, b, c\}$. Suppose A is listed in lexicographic order based on $0 < 1$ and B is in alphabetic order. If $A \times B \times A$ is listed in lexicographic order, then the next element after $((1, 0), c, (1, 1))$ is

A. $((1, 0), a, (0, 0))$
B. $((1, 1), c, (0, 0))$
C. $((1, 1), a, (0, 0))$
D. $((1, 1), a, (1, 1))$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

Which of the following statements is $\mathbf{TRUE}$?

A. For all sets A, B, and C, $A - (B - C) = (A - B) - C$.
B. For all sets A, B, and C, $(A - B) \cap (C - B) = (A \cap C) - B$.
C. For all sets A, B, and C, $(A - B) \cap (C - B) = A - (B \cup C)$.
D. For all sets A, B, and C, if $A \cap C = B \cap C$ then $A = B$.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

Which of the following statements is $\mathbf{FALSE}$?

A. $C - (B \cup A) = (C - B) - A$
B. $A - (C \cup B) = (A - B) - C$
C. $B - (A \cup C) = (B - C) - A$
D. $A - (B \cup C) = (B - C) - A$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

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